



Sound detection level

For

Sensor Lab mini project

**Third Year (Semester-VI) Bachelors in Information Technology, Autonomous
Program**

By

Student Name :Kartik Bhat

ClassD15ARoll no 03

Student Name :Spandan Deb

ClassD15ARoll no 13



Department of Information Technology
Vivekanand Education Society's Institute of Technology
An Autonomous Institute Affiliated to University of Mumbai
2023-2024

Certificate

This is to certify that **Kartik Bhat (03), Spandan Deb (13)** have completed the project report on the **Sound Level Detector Using Arduino** topic satisfactorily in partial fulfilment of the requirements for the award of Mini Project in sensor lab of third Year, (Semester-VI) in Information Technology under the guidance of Mr. Abhishek Chaudhari during the year 2024-2025 as prescribed by An Autonomous Institute Affiliated to University of Mumbai

Supervisor/ Examiner

Table of Contents

Chapter No.	Title	Page no.
	Abstract	4
	List of Figures	
	List of Tables	
Chapter 1:	Introduction	5
	1.1 Objective	
	1.2 Scope	
Chapter 2:	Literature Review	6
	2.1 Survey	
	2.2 Research Gap	
Chapter 3:	Project Description	7
	3.1 Problem Definition	
	3.2 Steps Involved	
	3.3 Block Diagram	
	3.4 Component Description/ Hardware & Software	
	3.5 Working of the project	
Chapter 4:	Implementation	10
	Hardware	
	Software (Flowchart/ Algorithms)	
Chapter 5:	Results	12
	References	13

Abstract

This project is based on detecting ambient sound levels using a low-cost, microcontroller-based solution. The system uses an Arduino Uno and an electret microphone module to detect variations in sound intensity and represents them visually using multiple LEDs. The project is designed for basic environmental noise level indication, educational demonstrations, and prototype sound monitoring systems. By using simple electronic components like resistors, capacitors, jumper wires, and a breadboard, the project becomes accessible to beginners and electronics hobbyists. This system can be expanded for real-time monitoring, alert systems, or as a basis for smart home integration.

Chapter 1

Introduction

1. Introduction

With the increasing concern for sound pollution and the growing need for interactive automation, sound detection systems have gained popularity. Devices that can interpret sound intensity are used in alarm systems, noise monitoring tools, interactive installations, and even voice-activated systems. The Arduino Uno, a popular open-source microcontroller platform, provides an excellent base for developing such systems in a cost-effective and simple manner.

1.2 Objectives

1. Identifying required components and their specifications.
2. Designing the circuit using a breadboard for easy prototyping.
3. Connecting the microphone module to Arduino's analog input pin.
4. Coding the Arduino to read analog values and control LEDs based on threshold levels.
5. Testing and calibrating the system for different sound intensities.
6. Optimizing the system for stable and responsive output.

Chapter 2

Review of Literature

1) Smart Embedded Framework using Arduino and IoT for Real-Time Noise and Air Pollution Monitoring and Alert system

The system integrates sensors for sound, gas, humidity, and temperature, connected to an Arduino microcontroller, with data transmitted via Wi-Fi to the Thing Speak cloud platform for remote access and analysis.

Testing in various pollution environments demonstrated effective monitoring and alerting capabilities, making it suitable for industrial areas, residential zones, hospitals, and regions with heavy traffic, while being cost-effective, efficient, and user-friendly.

2) An IoT Solution for Secured and Remote Sound Level Monitoring

The system employs a Raspberry Pi as the MQTT broker, facilitating data transfer between the publisher (Arduino with sound sensor) and subscriber (PC). The implementation ensures secure and efficient data communication, with the MQTT protocol enabling real-time monitoring and alerts.

3) The Working Principle of An Arduino.

The hardware components of the Arduino board, such as the USB plug, power supply, microcontroller, analog and digital pins, and reset button, are discussed in detail. The paper also covers the Arduino IDE (Integrated Development Environment), including its command area, text area for coding, and message window for feedback.

Practical examples, such as a sensor alarm system and a capacitance measuring circuit, are provided to illustrate how Arduino can be used in real-world projects. The paper concludes by summarizing the key aspects of Arduino's hardware and software, making it a comprehensive guide for beginners.

Chapter 3

Project Description

- **Problem Definition**

The absence of simple and affordable noise-level visualization systems limits experimentation and learning at the beginner level. Therefore, there is a need for a system that detects ambient sound levels and offers immediate visual feedback.

The electret microphone captures sound waves and converts them into electrical signals.

The Arduino reads these signals through its analog input pin and processes them. Based on predefined thresholds, the Arduino controls the LEDs to indicate the detected sound level.

The system can be calibrated using the Serial Monitor to adjust thresholds for optimal performance in different environments

- **Steps involved**

Identifying required components and their specifications.

Designing the circuit using a breadboard for easy prototyping.

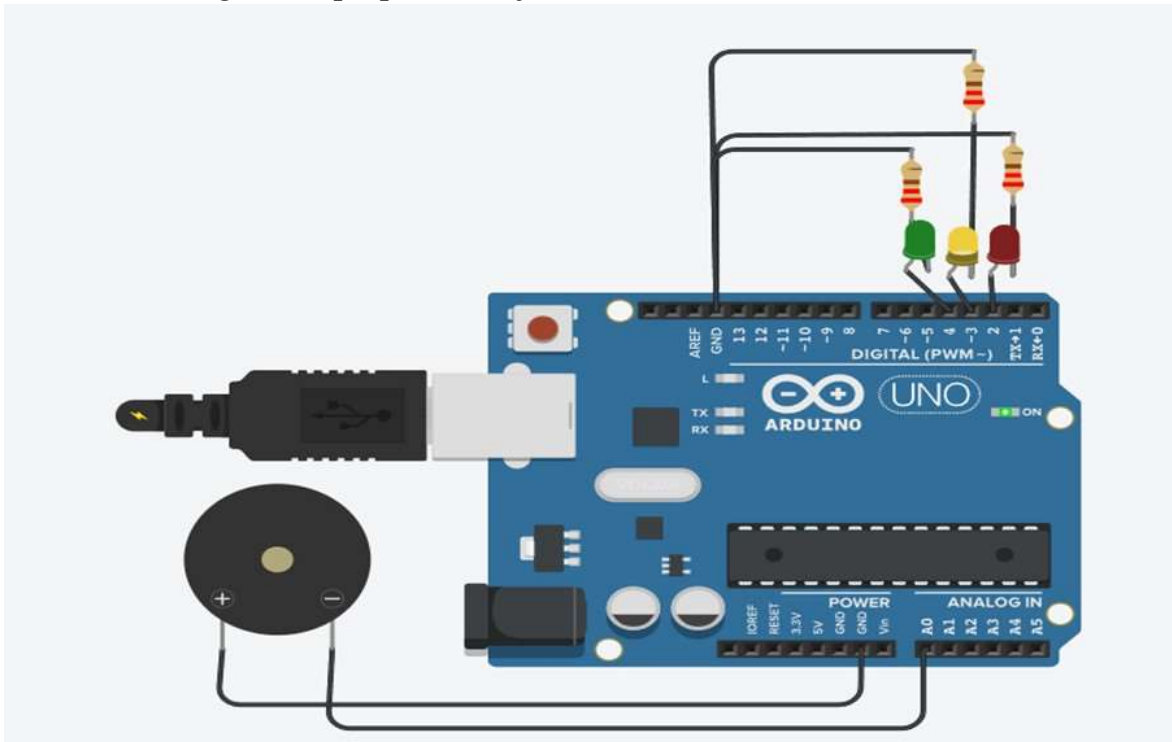
Connecting the microphone module to Arduino's analog input pin.

Coding the Arduino to read analog values and control LEDs based on threshold levels.

Testing and calibrating the system for different sound intensities.

Optimizing the system for stable and responsive output.

- Block Diagram of proposed Project

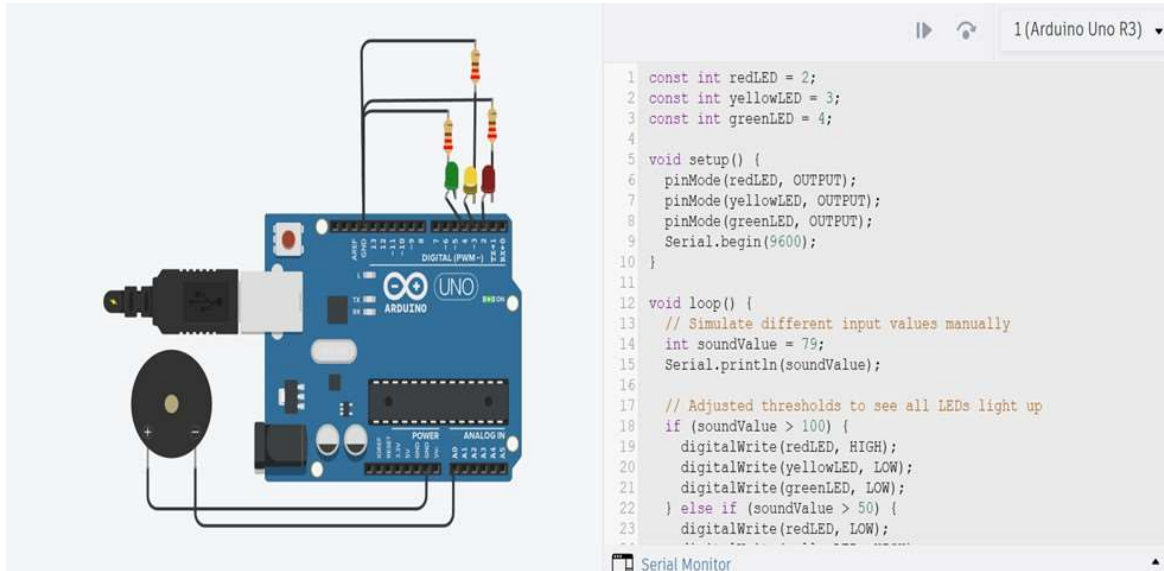


- Component Description

Component	Description
Arduino Uno	ATmega328P-based, 5V operating voltage, 16 MHz clock speed, 32 KB flash memory, 2 KB SRAM, 1 KB EEPROM, 14 digital I/O pins (6 PWM), 6 analog input pins
Electret Microphone	Condenser microphone with 20 Hz to 20 kHz frequency response, -44 dBV/Pa sensitivity, 2.2 k Ω impedance, operating at 3V to 10V DC.
Breadboard	Solderless prototyping board, sizes like 400 or 830 tie-points, plastic with conductive metal strips for temporary circuit assembly.

Resistors	Commonly 220Ω for LEDs, $\pm 5\%$ or $\pm 1\%$ tolerance, 1/4 watt or 1/2 watt power rating, carbon or metal film material.
Capacitors	Ceramic or electrolytic types, commonly 0.1 μF capacitance, voltage ratings from 16V to 50V, $\pm 5\%$ or $\pm 10\%$ tolerance for signal smoothing.
LEDs (5x)	Light Emitting Diodes in various colors, typically 1.8V to 3.3V forward voltage, 20 mA forward current, varying luminous intensity.
Jumper Wires	Male-to-male, male-to-female, or female-to-female types, typically copper with insulation, varying lengths for circuit connections.

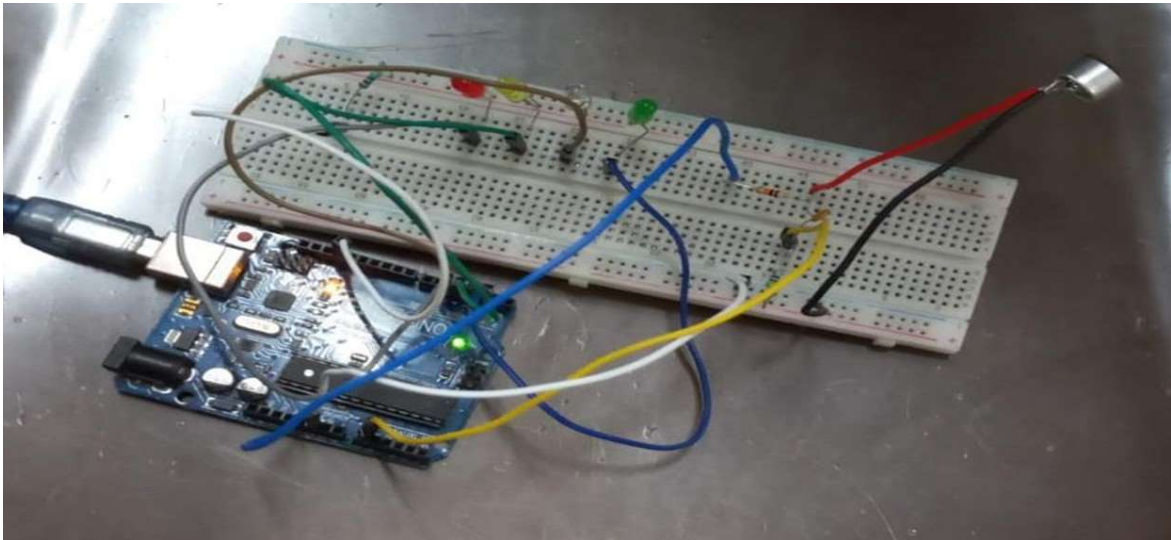
- **Working of proposed project**



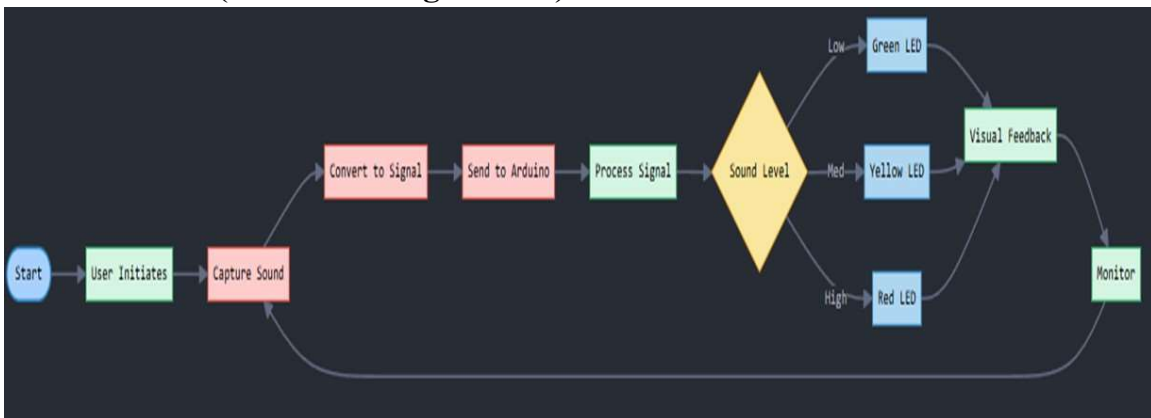
Chapter 4

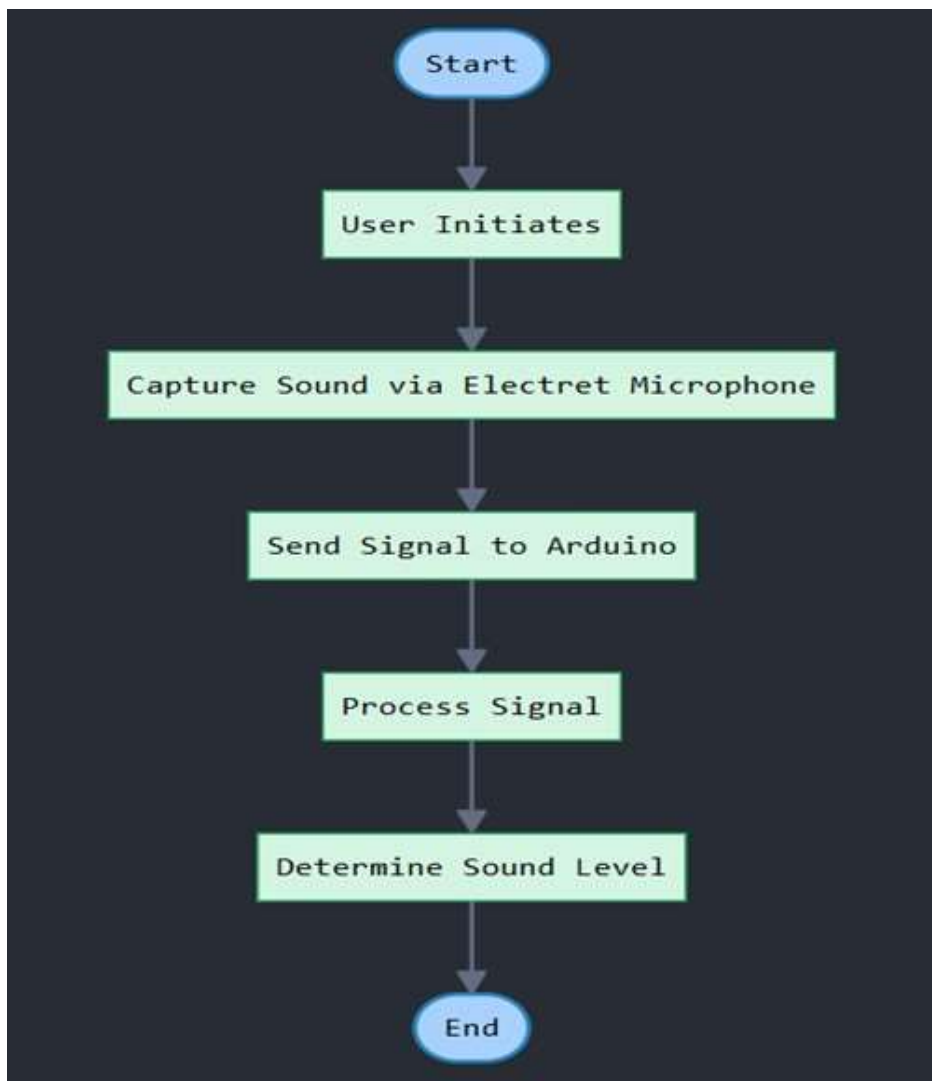
Implementation

- **Hardware**



- **Software (Flowchart/Algorithms)**





Chapter 5

Results

The electret microphone accurately captured varying sound levels and consistently generated analog signals across different environments.

The Arduino processed sound signals efficiently and triggered LEDs in real-time based on predefined thresholds.

LED indicators responded clearly and immediately, providing reliable visual feedback aligned with sound intensity.

The overall setup was stable and reliable across multiple tests, making it ideal for educational and DIY applications

Conclusion

Successfully developed a functional and interactive sound level detection system.

Integrated an electret microphone with Arduino for real-time monitoring and data collection.

Enabled accurate detection of varying sound intensities in different environments.

Showcased potential applications in noise pollution monitoring, smart environments, and educational demonstrations.

System is low-cost, easy to replicate, and ideal for DIY electronics and IoT projects.

Laid a strong foundation for future enhancements such as data logging, wireless transmission, and machine learning integration.

References

- [1] Saha, A. K., Sircar, S., Chatterjee, P., Dutta, S., Mitra, A., Chatterjee, A., ... & Saha, H. N. (2018, January). A raspberry Pi controlled cloud based air and sound pollution monitoring system with temperature and humidity sensing. In 2018 IEEE 8th Annual Computing and Communication Workshop and Conference (CCWC) (pp. 607-611). IEEE.
- [2] Nowshin, N., & Hasan, M. S. (2021, January). Microcontroller Based Environmental Pollution Monitoring System through IoT Implementation. In 2021 2nd International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST) (pp. 493-498). IEEE.
- [3] Morgan, J., & Shallcross, D. E. The power of sound—can we hear air pollution?.
- [4] Zhao, Y., Deng, G., Zhang, L., Di, N., Jiang, X., & Li, Z. (2021). Based investigate of beehive sound to detect air pollutants by machine learning. Ecological Informatics, 101246.